



Bank Initiated Transactions

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Object New **Bank to Many Wallet** operation
 and
 Bank Initiated Transactions

Author Vincent Cassotti

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Change history

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2 Document objective

This document presents several new features of the "Bank To Wallet" (B2W). The new **Bank2MultiWallet** (B2MW) operation, allowing a customer to transfer money from his bank account to the wallet of another Orange Money customer and the possibility to initiate transactions from the bank partner side, by exposing :

- a new API to the bank partner allowing to **initiate transfer operations** and **query their statuses**.
- a new API to the Mobile and USSD clients (Multibank) allowing to **list the pending operations** and **confirm their processing**.
- a new **USSD user journey** (Multibank) allowing to list and confirm operations pending operations.

Those new features will be available in the future releases **1.1.x** of the Multibank and the releases **1.0.x** of the B2W.

3 New BankToManyWallet operation (B2MW)

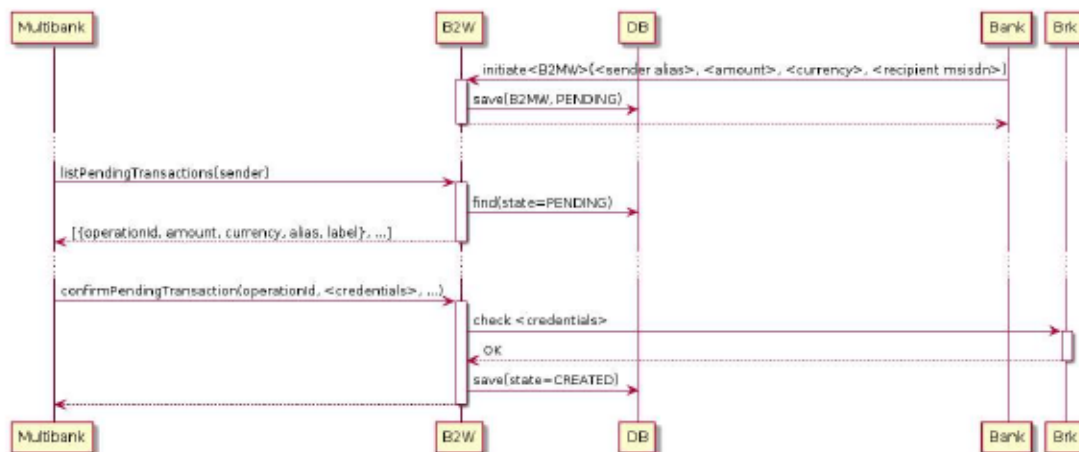
The new B2MW operation aims to let a user transfer money from his bank account to the wallet Orange Money of another user (only customers are allowed).

This new operation will be available on the new Bank API and its treatment will execute the following operations: to debit the bank account and to cash in the wallet.

1. **validate user's credential** (pin or pin/em)
2. **validate authorizations** for the future <cash in> operation
3. **debit the bank account of the originator**, which means to transfer money from his bank account to the main Orange Money bank account
4. **credit the Orange Money wallet of the recipient**, which means to execute a cash-in operation from the main wallet to the wallet of the recipient
5. **send two SMS** : one SMS to the originator and one to the recipient

3.1 Initiating a B2MW Transaction

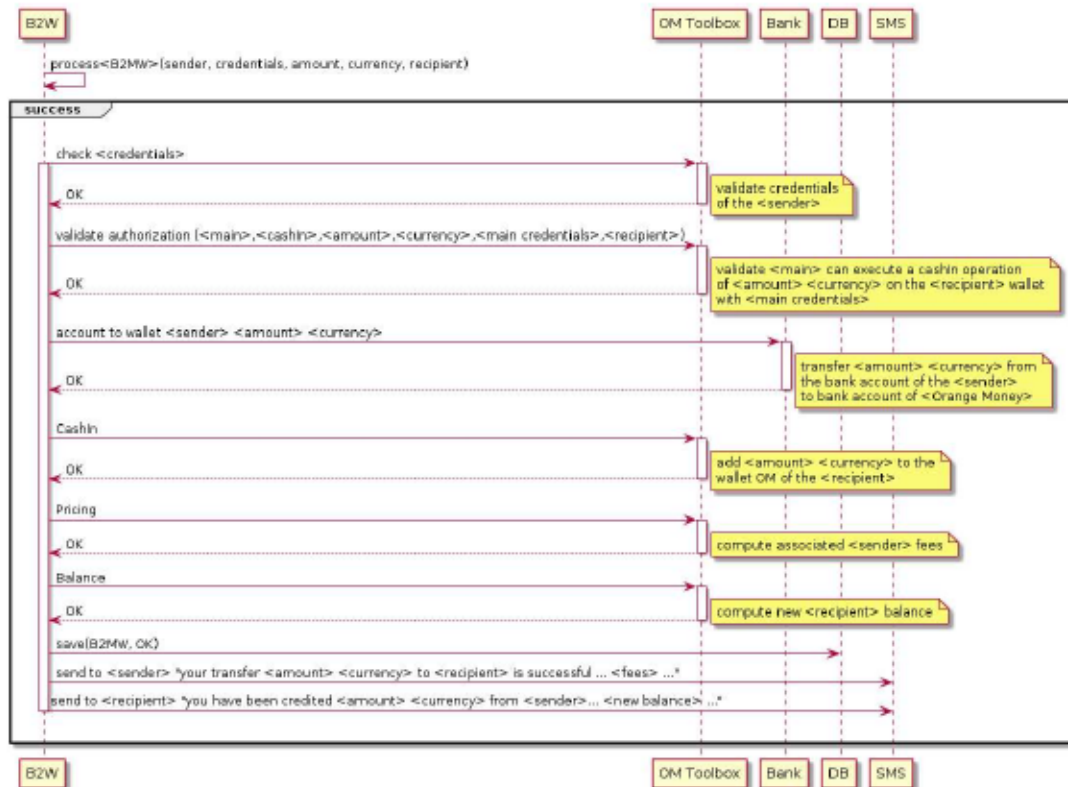
The operation will only be available from the Bank API. The sequence of operations will be more detailed in the corresponding section.



initiate and confirm a B2MW

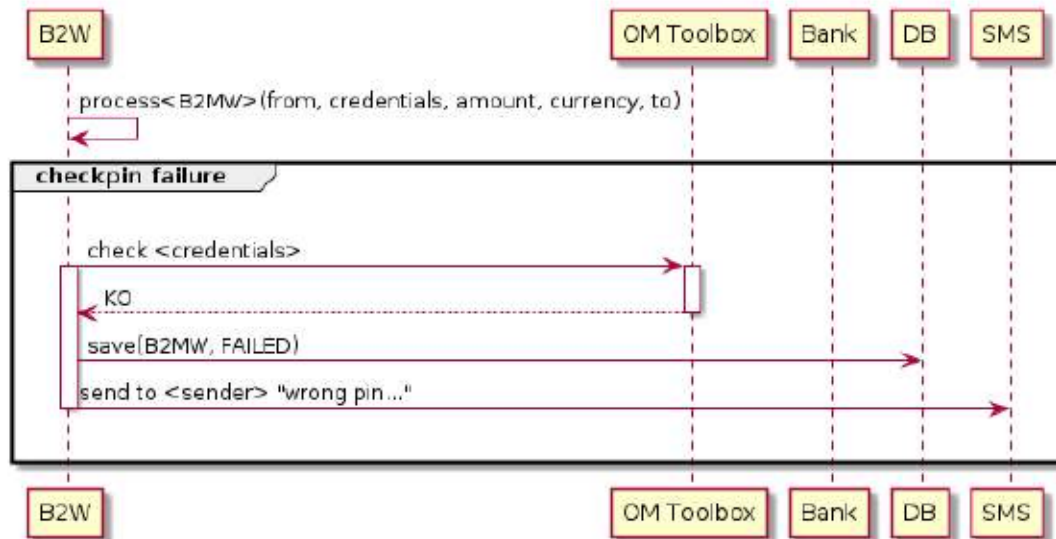
3.2 Processing of a B2MW Transaction

3.2.1 Success



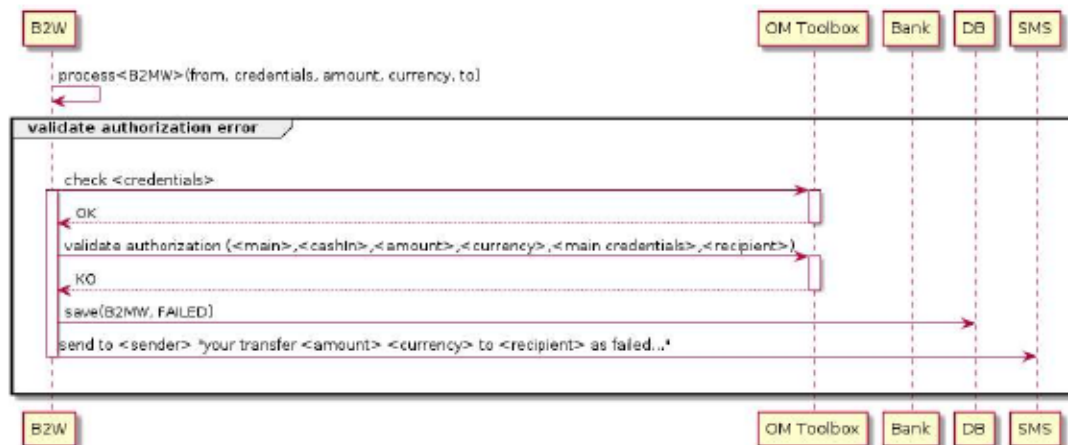
processing of a B2MW : success

3.2.2 Check pin failure



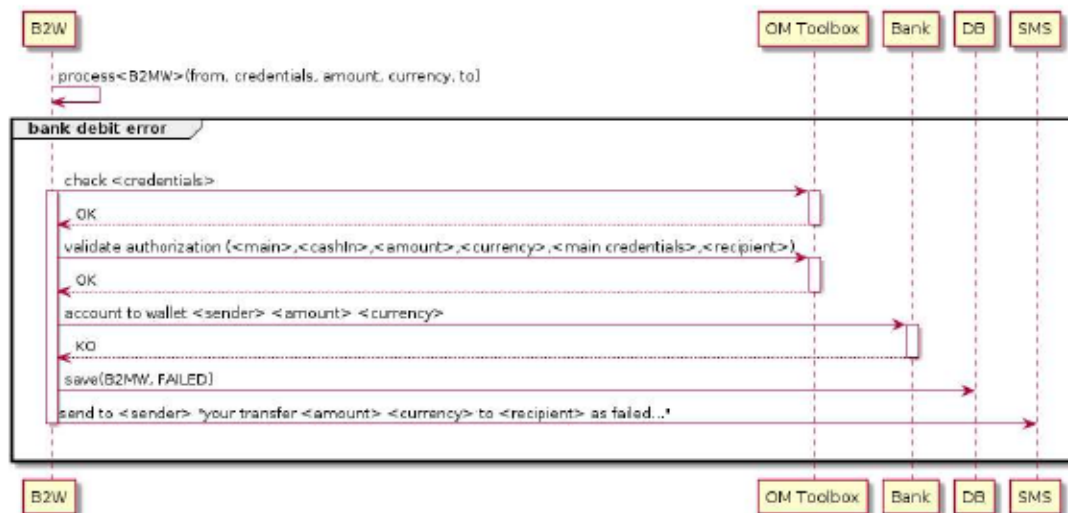
processing of a B2MW : check pin error

3.2.3 Check authorization failure



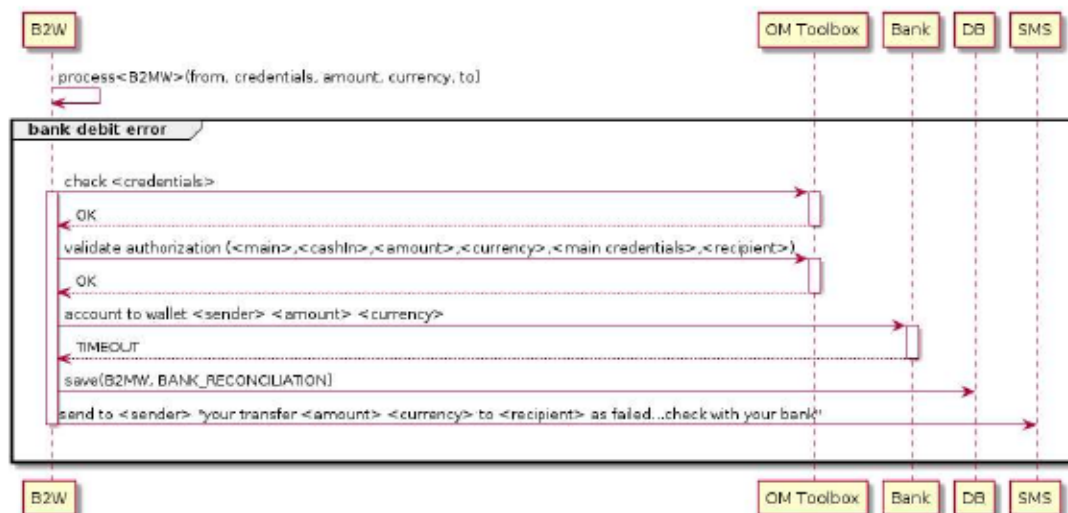
processing of a B2MW : validate authorization error

orange™ Business Services
3.2.4 Bank debit error



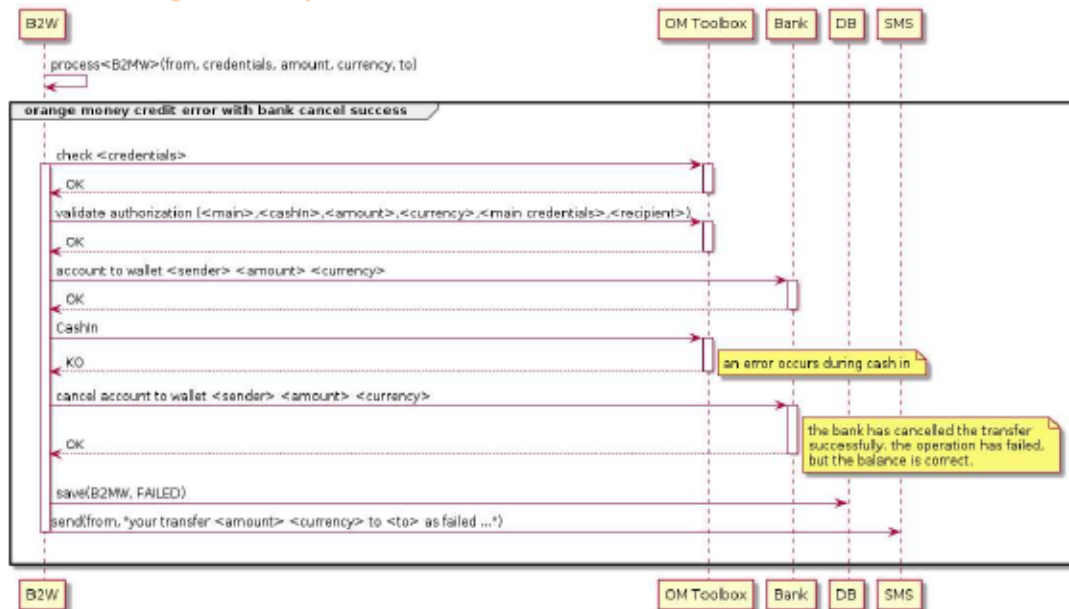
processing of a B2MW : bank debit error

3.2.5 Bank debit timeout



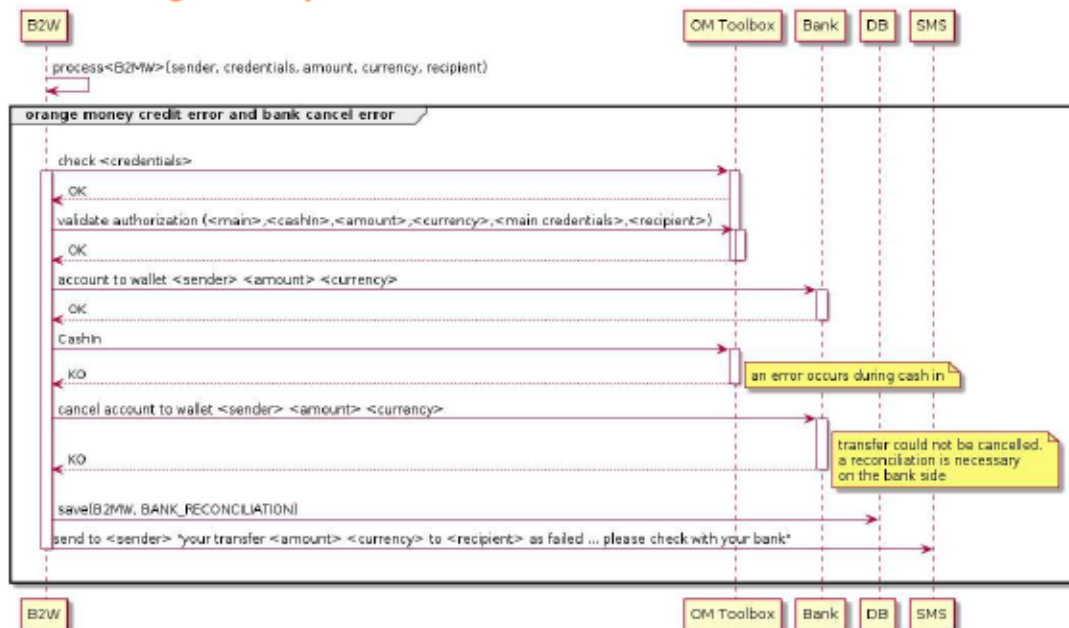
processing of a B2MW : bank debit error

3.2.6 Orange Money cash-in error



processing of a B2MW : orange money cash-in error

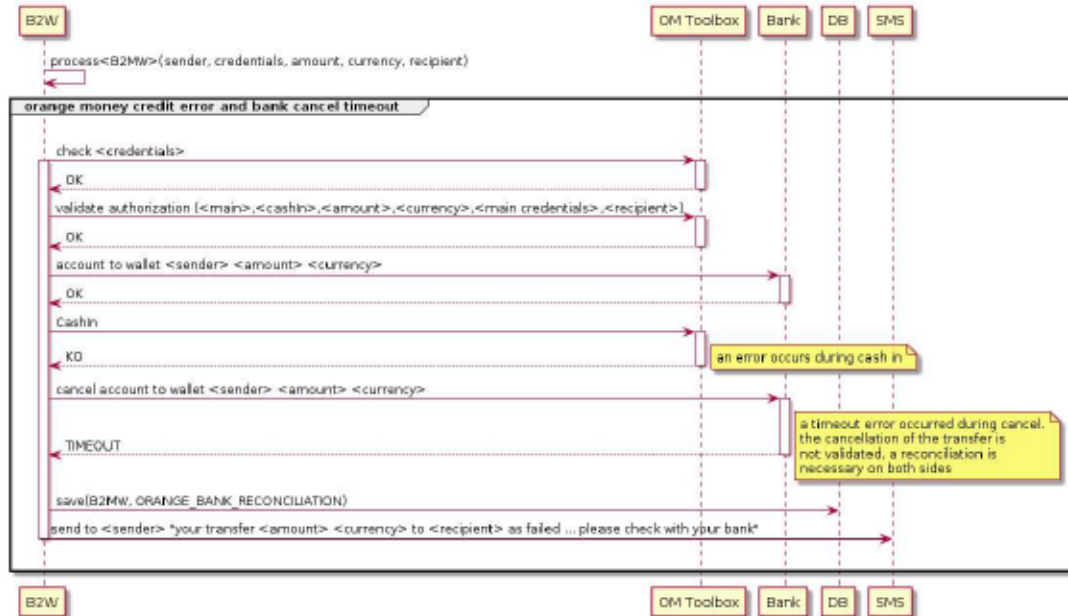
3.2.7 Orange Money cash-in error with bank cancel error



processing of a B2MW : orange money cash-in error followed by bank cancel error



3.2.8 Orange Money cash-in error with bank cancel timeout



processing of a B2MW : orange money cash-in error followed by bank cancel timeout



4 New Bank API (B2W)

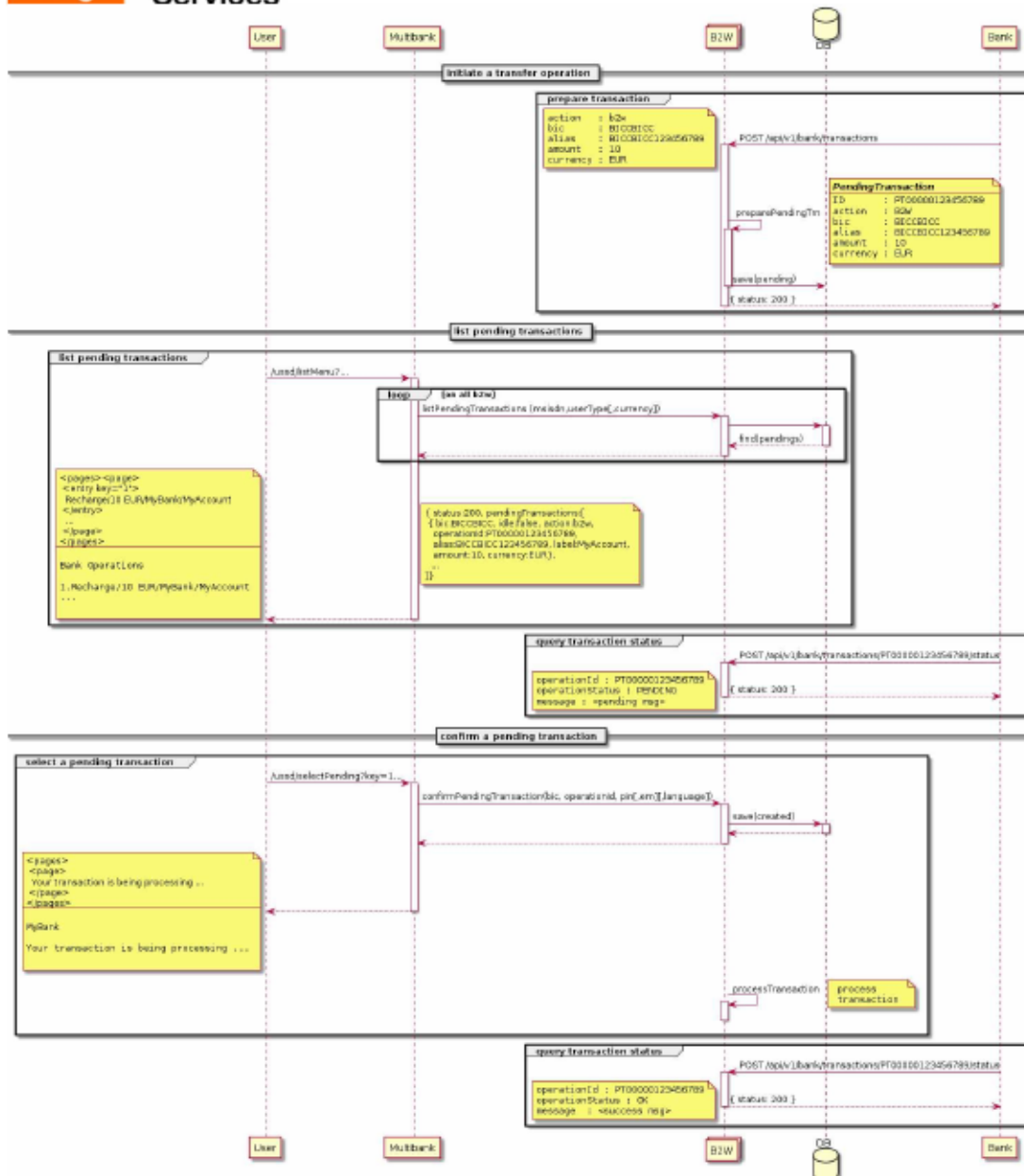
Using this new API, Bank partners are now allowed to ***initiate a transfer operation*** for their users :

- from a wallet Orange Money to the associated bank account of the user,
- from a bank account to the associated wallet Orange Money of the user,
- and from a bank account to the wallet of another Orange Money customer.

The operation will be in a pending state, and the owner of the bank account will have to confirm the operation in order to really start its processing.

Once the operation is confirmed, the service behaves exactly as if the operation had been initiated normally.

The bank partner can also query the B2W and retrieve the ***transaction status*** about previously initiated operations.



bank initiated transaction sequence diagram



4.1 Initiating a transfer operation

4.1.1 Type definition

bank API initiateTransaction input parameters

Name	Type	Mandatory	Description
action	String	yes	Type of the operation w2b, b2w, b2mw
alias	String	yes	Bank account identifier of the transfer originator
amount	String	yes	Amount of the transfer
language	String	No	Country code (ISO 3166) fr, ml, ci ... This parameter is optional, and will be used to filter pending transactions by currency
to	Recipient	No ₁	Recipient of the transfer operation

bank API subtype Recipient

Name	Type	Mandatory	Description
msisdn	String	Yes	Phone number of the recipient
userType	String	Yes	Type of user customer or retailer

bank API initiateTransaction output results

Name	Type	Mandatory	Description
operationId	String	Yes	Status of the request

bank API initiateTransaction output results on error

Name	Type	Mandatory	Description
code	String	Yes	error code
message	String	Yes	error message

4.1.2 API Description

bank initiateTransaction request url.

POST /api/v1/bank/transactions

Content-Type: application/json



bank initiateTransaction request body.

```
{
  "action": "b2mw" 1,
  "alias": "ECOCMLBA62364311285901",
  "amount": "100",
  "language": "en" 2,
  "to": {
    "msisdn": "0037345811",
    "userType": "customer" 3
  }
}
```

¹ The type of the operation : *w2b, b2w, b2mw*

² Optional, default language will be used, used for sms translation

³ The only authorized value is *customer*

bank initiateTransaction response with status 200 : operation has been initiated successfully.

```
{
  "operationId": "MBXXX.YYYYMMDDhhmmss.RND"
}
```

.bank initiateTransaction response with status 400 : a parameter is not valid (empty, invalid, ...).

```
{
  "code": "<error code>",
  "message": "<error message>"
}
```

.bank initiateTransaction response with status 500 : an error has been encountered during the initiation of the operation.

```
{
  "code": "<error code>",
  "message": "<error message>"
}
```

4.2 Knowing the status of an operation

4.2.1 Type definition

bank API transactionStatus input parameters

Name	Type	Mandatory	Description
operationId	String	yes	Type of the operation w2b, b2w, b2mw

bank API transactionStatus output results

Name	Type	Mandatory	Description
operationId	String	Yes	ID of the operation
operationStatus	String	Yes	Status of the operation
message	String	Yes	message to the user

bank API transactionStatus output results on error

Name	Type	Mandatory	Description
message	String	Yes	error message

4.2.2 API Description

bank transactionStatus request url.

GET /api/v1/bank/transactions/{operationId}/status

bank transactionStatus response with status 200 : transaction status has been successfully retrieved.

```
{
  "operationId" : <operation identifier>,
  "operationStatus" : <operation status>,
  "message" : <message associated to the current transaction state/status>
}
```

bank transactionStatus response with status 404 : transaction identifier not found.

// no response body

bank transactionStatus response with status 500 : an error occurred when retrieve transaction status.

```
{
  "message" : "<error message>"
}
```




Status	Description
PENDING	the operation was initiated by the bank but not yet confirmed by the user
EXPIRED	the operation was not confirmed on time by the user
PROCESSING	the operation is being processed
OK	operation processing has been successfully completed
KO	the processing of the operation has failed
BANK_RECONCILIATION	the processing of the operation ended in failure with the need to reconcile on the bank side
ORANGE_MONEY_RECONCILIATION	the processing of the operation ended in failure with the need to reconcile on the Orange Money side
ORANGE_BANK_RECONCILIATION	the processing of the operation ended in failure with the need to reconcile on both sides

message

State of the operation	Description
operation is not yet completed or expired	PENDING, PROCESSING, EXPIRED The message key is based on the processing state of the transaction (the status of the transaction is not used) • messageForBankStatePENDING • messageForBankStatePROCESSING • messageForBankStateEXPIRED
operation not already completed	OK, KO, BANK_RECONCILIATION, BANK_RECONCILIATION, ORANGE_BANK_RECONCILIATION The message is based on the internal status of the transaction in the same way that SMS messages are built. It is possible to specialize the message key associated with each internal status in configuration.



If the associated status does not exist in configuration, the message is built from a specific key ***messageForBankUNDEFINED*** (should not be possible since configuration validation would raise an error)

Note

Replacement variables can be used in messages. Their Values are defined using the information saved in the transaction.



5 Mobile API (Multibank)

The new bank API creates transactions with the processing state **PENDING**.

Multibank clients can :

- query those transactions by calling the service **listPendingTransactions**.
- confirm their processing by calling the service **confirmPendingTransaction**.

5.1 List pending transactions

This service will return information about the current PENDING transactions from all associated B2W.

5.1.1 Type definition

mobile listPendingTransactions input parameters

Name	Type	Mandatory	Description
msisdn	String	yes	Phone number of the user
userType	String	yes	Type of user customer or retailer
countryCode	String	yes	Country code (ISO 3166) fr, ml, ci ...
currency	String	No	Currency of the transfer (ISO 4217-3) EUR, XOF, USD ... This parameter is optional, and will be used to filter pending transactions by currency

mobile listPendingTransactions output results

Name	Type	Mandatory	Description
status	String	Yes	Status of the request
pendingTransactions	Transaction[]	Yes	Array of pending transactions



Name	Type	Mandatory	Description
bic	String	Yes	Bank identifier
bankLabel	String	Yes	Bank Name
idle	Boolean	Yes	Bank Idle State
action	String	Yes	Type of the operation
operationId	String	Yes	ID of the operation
amount	String	Yes	Amount of the transfer
currency	String	Yes	Currency of the transfer (ISO 3166) EUR, XOF, USD ...
alias	String	Yes	Bank account alias of the originator
label	String	Yes	Bank account label of the originator
to	Recipient	No	Only present for a B2MW, defines the target wallet information

mobile returned subtype Recipient

Name	Type	Mandatory	Description
msisdn	String	Yes	Phone number of the recipient
userType	String	Yes	Type of user customer or retailer



5.1.2 API Description

POST /mobile

Content-Type: application/json

mobile listPendingTransactions jsonrpc request.

```
{
  "jsonrpc": "2.0",
  "method": "mobilebank.listPendingTransactions",
  "params": {
    "msisdn": "70812345",
    "userType": "customer",
    "countryCode": "ml",
    "currency": "XOF" 1
  },
  "id": "<id>"
}
```

¹ Optional

mobile listPendingTransactions jsonrpc response.

```
{
  "jsonrpc": "2.0",
  "result": {
    "status" : 200,
    "operations": [
      {
        "bic" : "ECOCMLBA",
        "bankLabel": "Bank Label",
        "idle" : false,
        "action": "b2mw",
        "operationId" : "MBRAW.20210921181747.983",
        "amount" : "2000",
        "currency" : "XOF",
        "alias" : "ECOCMLBA62364311285901",
        "label" : "Bank Account Label",
        "to": { 1
          "msisdn" : "70854321",
          "userType": "customer"
        }
      }
    ]
  },
  "id": "<id>"
}
```

¹ Only present for B2MW operation



5.2 Confirm a pending transaction

5.2.1 Type definition

mobile confirmPendingTransaction input parameters

Name	Type	Mandatory	Description
bic	String	yes	Bank identifier of the operation to confirm
operationId	String	yes	Identifier of the operation to confirm
pin	String	yes	PIN Code of the user
em	String	yes	EM of the user
language	String	no	Country Code (ISO 3166) <i>fr, ml, ci ...</i> Optional, used to translate messages

mobile confirmPendingTransaction output results

Name	Type	Mandatory	Description
status	Integer	Yes	Status of the request
operationId	String	Yes	Identifier of the operation to confirm
brokerMapCode	String	Yes	Broker Mapping Code

5.2.2 API Description

POST /mobile

Content-Type: application/json

mobile confirmPendingTransaction jsonrpc request.

```
{
  "jsonrpc": "2.0",
  "id": "<id>",
  "method": "mobilebank.confirmPendingTransaction",
  "params": {
    "bic": "ECOCMLBA",
    "operationId": "MBRCH.20190122094355.876",
    "pin": "1234",
    "em": "c2VjcmV0IHZhbHVlCg",
    "language": "en" 1
  }
}
```

¹ Optional

mobile confirmPendingTransaction jsonrpc response.

```
{
  "jsonrpc": "2.0",
  "id": "<id>",
  "result": {
    "status": 200,
    "operationId": "MBRCH.20190122094355.876",
    "brokerMapCode": "SUCCESS"
  }
}
```




6 USSD API (Multibank)

The new bank API creates transactions with the processing state **PENDING**.

Multibank clients can :

- query those transactions by calling the service **listPendingTransactions**.
- confirm their processing by calling the service **confirmPendingTransaction**.

6.1 List pending transactions

This service will return information about the current PENDING transactions from all associated B2W,

6.1.1 Type definition

ussd listPendingTransactions input parameters

Name	Type	Mandatory	Description
msisdn	String	yes	Phone number of the user
userType	String	yes	Type of user <i>customer</i> or <i>retailer</i>
countryCode	String	yes	Country code (ISO 3166) <i>fr, ml, ci ...</i>
currency	String	No	Currency of the transfer (ISO 4217-3) <i>EUR, XOF, USD ...</i> This parameter is optional, and will be used to filter pending transactions by currency

ussd listPendingTransactions output results

Name	Type	Mandatory	Description
status	String	Yes	Status of the request
pendingTransactions	Transaction[]	Yes	Array of pending transactions

subtype Transaction

Name	Type	Mandatory	Description
bic	String	Yes	Bank identifier
bankLabel	String	Yes	Bank label
idle	Boolean	Yes	Bank Idle State
action	String	Yes	Type of the operation



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operationId	String	Yes	ID of the operation
amount	String	Yes	Amount of the transfer
currency	String	Yes	Currency of the transfer (ISO 3166) EUR, XOF, USD ...
alias	String	Yes	Bank account alias of the originator
label	String	Yes	Bank account label of the originator
to	Recipient	No	Only present for a B2MW, defines the target wallet information

subtype Recipient

Name	Type	Mandatory	Description
msisdn	String	Yes	Phone number of the recipient
userType	String	Yes	Type of user customer or retailer

6.1.2 Api Description

POST /ussd

Content-Type: application/xml

ussd listPendingTransactions xmlrpc request.

```
<?xml version="1.0" encoding="utf-8"?>
<methodCall>
  <methodName>mbank.listPendingTransactions</methodName>
  <params>
    <param>
      <value>
        <struct>
          <member>
            <name>msisdn</name>
            <value>
              <string>70812345</string>
            </value>
          </member>
          <member>
            <name>userType</name>
            <value>
              <string>customer</string>
            </value>
          </member>
          <member>
            <name>countryCode</name>
            <value>
```



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```
<string>ml</string>
</value>
</member>
<member>
  <name>currency</name>
  <value>
    <string>XOF</string>
  </value>
</member>
</struct>
</value>
</param>
</params>
</methodCall>
```

ussd listPendingTransactions xmlrpc response.

```
<?xml version="1.0" encoding="UTF-8"?>
<methodResponse>
  <params>
    <param>
      <value>
        <struct>
          <member>
            <name>status</name>
            <value>
              <string>200</string>
            </value>
          </member>
          <member>
            <name>pendingTransactions</name>
            <value>
              <array>
                <data>
                  <value>
                    <struct>
                      <member>
                        <name>bic</name>
                        <value>
                          <string>ECOCMLBA</string>
                        </value>
                      </member>
                      <member>
                        <name>bankLabel</name>
                        <value>
                          <string>Bank Label</string>
                        </value>
                      </member>
                      <member>
```



```
<name>idle</name>
<value>
  <boolean>0</boolean>
</value>
</member>
<member>
  <name>action</name>
  <value>
    <string>b2mw</string>
  </value>
</member>
<member>
  <name>operationId</name>
  <value>
    <string>MBRCH.20210914120000.976</string>
  </value>
</member>
<member>
  <name>amount</name>
  <value>
    <string>10</string>
  </value>
</member>
<member>
  <name>currency</name>
  <value>
    <string>EUR</string>
  </value>
</member>
<member>
  <name>alias</name>
  <value>
    <string>ECOCMLBA62364311285901</string>
  </value>
</member>
<member>
  <name>label</name>
  <value>
    <string>Account Label</string>
  </value>
</member>
<member>
  <name>to</name>
  <value>
    <struct>
      <member>
        <name>msisdn</name>
        <value>
          <string>70854321</string>
        </value>
      </member>
    </struct>
  </value>
</member>
```



```
        </value>
      </member>
      <member>
        <name>userType</name>
        <value>
          <string>customer</string>
        </value>
      </member>
    </struct>
  </value>
</member>
</struct>
</value>
</data>
</array>
</value>
</member>
</struct>
</value>
</param>
</params>
</methodResponse>
```

¹ Only present for B2MW operation



6.2 Confirm a pending transaction

6.2.1 Type definition

ussd confirmPendingTransaction input parameters

Name	Type	Mandatory	Description
bic	String	yes	Bank identifier of the operation to confirm
operationId	String	yes	Identifier of the operation to confirm
pin	String	yes	PIN Code of the user
language	String	no	Country Code (ISO 3166) <i>fr, ml, ci</i> ... Optional, used to translate messages

ussd confirmPendingTransaction output results

Name	Type	Mandatory	Description
status	String	Yes	Status of the request
message	String	Yes	Message to the user

6.2.2 Api Description

POST /ussd

Content-Type: application/xml

```
<?xml version="1.0" encoding="utf-8"?>
<methodCall>
<methodName>mbank.confirmPendingTransaction</methodName>
<params>
  <param>
    <value>
      <struct>
        <member>
          <name>bic</name>
          <value>
            <string>ECOCMLBA</string>
          </value>
        </member>
        <member>
          <name>operationId</name>
          <value>
            <string>MBRCH.20210914120000.976</string>
          </value>
        </member>
        <member>
          <name>pin</name>
```



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```
<value>
  <string>1234</string>
</value>
</member>
<member>
  <name>language</name>
  <value>
    <string>fr</string>
  </value>
</member>
</struct>
</value>
</param>
</params>
</methodCall>
```

ussd confirmPendingTransaction xmlrpc response.

```
<?xml version="1.0" encoding="utf-8"?>
<methodResponse>
  <params>
    <param>
      <value>
        <struct>
          <member>
            <name>status</name>
            <value>
              <string>301</string>
            </value>
          </member>
          <member>
            <name>message</name>
            <value>
              <string>[Name_Banque_selectionnée] Votre demande a bien été
                prise en compte, vous allez recevoir un SMS avec le solde ban
                caire de votre
                compte [account_label]
              </string>
            </value>
          </member>
        </struct>
      </value>
    </param>
  </params>
</methodResponse>
```


7 USSD Journey (Multibank)

New USSD user's journey will have a new menu entry to the *Bank Operations* screen.



pending transactions USSD user's journey

Bank Operations

adds a new menu to access **Display Pending Transaction Screen**

Operations bancaires

Selectionnez votre banque:

1. ABU
2. Ecobank
3. <List_pending_transactions>

Display pending transactions

displays a list of menu entries describing pending transactions, using the following replacements :

- %BANK% (Bank name)
- %ALIAS% (Account alias)
- %LABEL% (Account label)
- %AMOUNT% (Amount)
- %CURRENCY% (Currency)
- %IDLE% (Idle state)
- %RECIPIENT% (Msisdn of the recipient - B2MW only)



then the user selects an entry to access **Summary screen**

Note

Some replacements variables have double %, which means that their placeholders will be replaced by the replacement of their values. For example, given that the variable CURRENCY has value XOF, %%CURRENCY%% will be replaced by the translation of the message key XOF to be found in the translation file.

Operations bancaires

<Select_your_transaction>

1. <pendingB2W>
2. <pendingW2B>
3. <pendingB2MW>

Note

if the bank is in idle state, a special **Idle State Warning** screen is displayed and the user can choose to cancel or to continue.

Summary

describes the selected operation and invites the user to enter his pin code.

<bank>

<transfer_[b2w|w2b|b2mw]_summary>

if the code is entered correctly, the transaction is processed and the **Confirmation** Screen is displayed.

if the code is not entered correctly, the **Retry Pin** screen is displayed. On the third error, the transaction is aborted and the **Transaction Aborted** screen is displayed.

Confirmation

display the message associated to the confirmation process.

message translations

Message key	English translation (en)	French translation (fr)
List_pending_transactions	Pending transactions	Transactions en attente
Select_your_transaction	Select your transaction:	Transactions en attente:
pendingB2W	Top up %BANK%%IDLE%% %LABEL% %AMOUNT% %%CURRENCY%%	Recharge %BANK%%IDLE%% %LABEL% %AMOUNT% %%CURRENCY%%
pendingW2B	Unload %BANK%%IDLE%%	Decharge %BANK%%IDLE%%



	%LABEL% %AMOUNT% %%CURRENCY%%	%LABEL% %AMOUNT% %%CURRENCY%%
pendingB2MW	Transfer %BANK%%IDLE%% %LABEL% %AMOUNT% %%CURRENCY%% to %RECIPIENT%	Transfert %BANK%%IDLE%% %LABEL% %AMOUNT% %%CURRENCY%% a %RECIPIENT%
is_idle	*	*
is_not_idle		
transfer_b2w_summary	You have requested a transfer of %AMOUNT% %%CURRENCY%% from %LABEL% to Orange Money. To confirm, please enter your Orange Money secret code	Transfert de %AMOUNT% %%CURRENCY%% de %LABEL% vers Orange Money. Pour confirmer veuillez saisir votre code secret Orange Money
transfer_w2b_summary	You have requested a transfer of %AMOUNT% %%CURRENCY%% from Orange Money to %LABEL%. To confirm, please enter your Orange Money secret code	Transfert de %AMOUNT% %%CURRENCY%% de Orange Money vers %LABEL%. Pour confirmer veuillez saisir votre code secret Orange Money
transfer_b2mw_summary	You have requested a transfer of %AMOUNT% %%CURRENCY%% from %LABEL% to the wallet Orange Money of %RECIPIENT%. To confirm, please enter your Orange Money secret code	Transfert de %AMOUNT% %%CURRENCY%% de %LABEL% vers le compte Orange Money du %RECIPIENT%. Pour confirmer veuillez saisir votre code secret Orange Money

8 Updates of the Customer Care

8.1 New Transaction B2MW

<new transaction detailed view for the B2MW> ...

8.2 New Transaction states

<new transaction list view showing B2MW, PENDING and EXPIRED transactions > ...

<new transaction detailed view PENDING and EXPIRED transaction > ...



**Business
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9 Impacts on the Global Reporting and Daily Events Files (FQE)

Global Reporting as well as the Event Daily File generation are based on the unit operations carried out on the partner side and Orange Money. Adding a new transaction will have no impact on the generation of FQE files and GR reports.

Note

It will, however, be necessary to ensure that the processing of the transaction will record the correct events and that the generated files and the reporting requests generate the data associated with the new operation.